# Water Testing Kit



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

**Covers these lab items:** Water testing kit

## What it is used for

A water testing kit measures selected physical or chemical parameters in a water sample using test strips, reagents, meters, or comparison charts. Correct sampling and test procedures help produce reliable results for water-quality screening.

## Main parts

- Sample collection bottle or container
- Test strips, reagent bottles, or test tablets
- Colour comparison chart or digital meter
- Sample tubes, cuvettes, or measuring accessories
- Dropper, scoop, or measuring syringe
- Instruction sheet and result record form

## Before you start

1. Read the kit instructions and identify the parameters and measurement ranges it supports.
2. Check expiry dates, reagent condition, meter battery, and calibration requirements.
3. Use clean sample containers suitable for the parameter being tested.
4. Collect a representative sample using the prescribed sampling method and avoid touching the inside of the container.
5. Label the sample with its source, location, date, and collection time.
6. Wear suitable gloves and eye protection when handling reagents, and review their safety information.

## How to use

1. Allow the sample to reach the temperature required by the kit instructions if necessary.
2. Measure the specified sample volume using the supplied or approved measuring accessory.
3. Add the correct test strip, tablet, or reagent quantity in the required order.
4. Mix or wait for the reaction for the exact time specified by the manufacturer.
5. For strip tests, compare the colour within the stated reading window and under suitable lighting.
6. For digital meters, calibrate as required, rinse the probe, and immerse it to the specified depth.
7. Record the displayed value or colour result with the correct units and parameter name.
8. Repeat the test when quality-control instructions require confirmation or the result appears abnormal.
9. Dispose of used strips, reagents, and samples according to the kit instructions and laboratory rules.

10. Clean the equipment, close reagent containers, and store the kit under the recommended conditions.

## ⚠ Safety

- Never taste or directly smell a water sample or chemical reagent.
- Wear gloves and eye protection when the reagent safety information requires them.
- Do not mix reagents from different kits or use expired chemicals.
- Keep reagents away from children, food, and drinking-water containers.
- Follow laboratory procedures for handling samples suspected of containing biological or chemical hazards.
- Do not assume a screening kit proves water is safe for drinking; use an approved laboratory when required.

## Care & storage

- Keep test strips and reagents sealed and protected from heat, moisture, and sunlight.
- Rinse probes and sample tubes with the recommended water after use.
- Calibrate digital meters and replace batteries according to the manufacturer's schedule.
- Record reagent lot numbers, expiry dates, calibration, and quality-control results where required.

## Common mistakes to avoid

- Using contaminated containers can change the measured result.
- Reading a colour strip too early or too late can produce an incorrect value.
- Using expired reagents or an uncalibrated meter can reduce accuracy.
- Recording results without units or sample identification can make them unreliable or unusable.

*Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.*